Rear wheel: simulation versus 50 Hz hardware replay

Same 50 Hz-held model_2500 torque sequence · source episode 12.5–17.5 s
Airborne hardware versus saved nominal fixed-base simulation · actual timestamps, no time shift

Actuator torque before passive friction

Sim min/max: -0.233 / 0.009 Hardware min/max: -0.218 / 0.012

Torque (Nm)

Isaac Lab 400 Hz — submitted actuator effort
Hardware replay, C0 50 Hz — A1 current × Kt

Rear-wheel velocity

Sim min/max: -21.398 / 0.000 Hardware min/max: -8.049 / 0.099

Angular velocity (rad/s)

Isaac Lab 400 Hz
Hardware replay, C0 50 Hz

Cumulative rear-wheel displacement

Sim min/max: -33.290 / 0.000 Hardware min/max: -5.283 / 0.000

Relative position (rad)

Isaac Lab 400 Hz
Hardware replay, C0 50 Hz

Time since recorded torque sequence begins (s)

Torque is actuator-level: simulation effort versus current-derived hardware estimate, not net torque after losses.
Position preserves full turns relative to each run’s initial reference; support and actuator dynamics are not fully matched.